

Design Report 3

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EML2322L – Spring 2026

4/12/2026

Concept Description

This design is a rotational blade food slicer that is powered by dual-action two-stroke compressed air powered piston engine. To connect to the air engine, the food slicer uses a pulley system. In this system, a pulley flat belt (OTS6) made of Buna-N rubber & Nylon plastic sits on the outer cylindrical face of the flywheel and the indented surface of the shaft (P5). The small indent in the shaft diameter guarantees that the pulley flat belt cannot freely move up and down the shaft. As the flywheel rotates, the pulley flat belt causes the shaft to rotate, therefore allowing for the cutting motion. The shaft is made of 6061 aluminum because of its low cost and ease of machinability. Along the shaft, there is a blade holder (P7) with 2 blades (P6) attached to it. The blade holder is necessary because it provides a flat surface that the blades can then be fastened to. The blade holder is made of 6061 aluminum again because of its low cost and ease of machinability. The blades are made of 304 stainless steel because they need to be strong enough to operate at high speeds and to cut through various foods. 304 stainless steel is a better choice than aluminum because aluminum is softer and more likely to deflect in this scenario. The blades are also created using waterjet cutting to get the specific radius necessary for the blades. The shaft, blade holder, and blades are fastened together with a total of four 0-80 screws made of 18-8 stainless steel (OTS5), with two screws being used for each blade. Considering the small radius of the shaft, the high thread per inch count on the screws ensures that there will be 5 threads of engagement in the shaft.

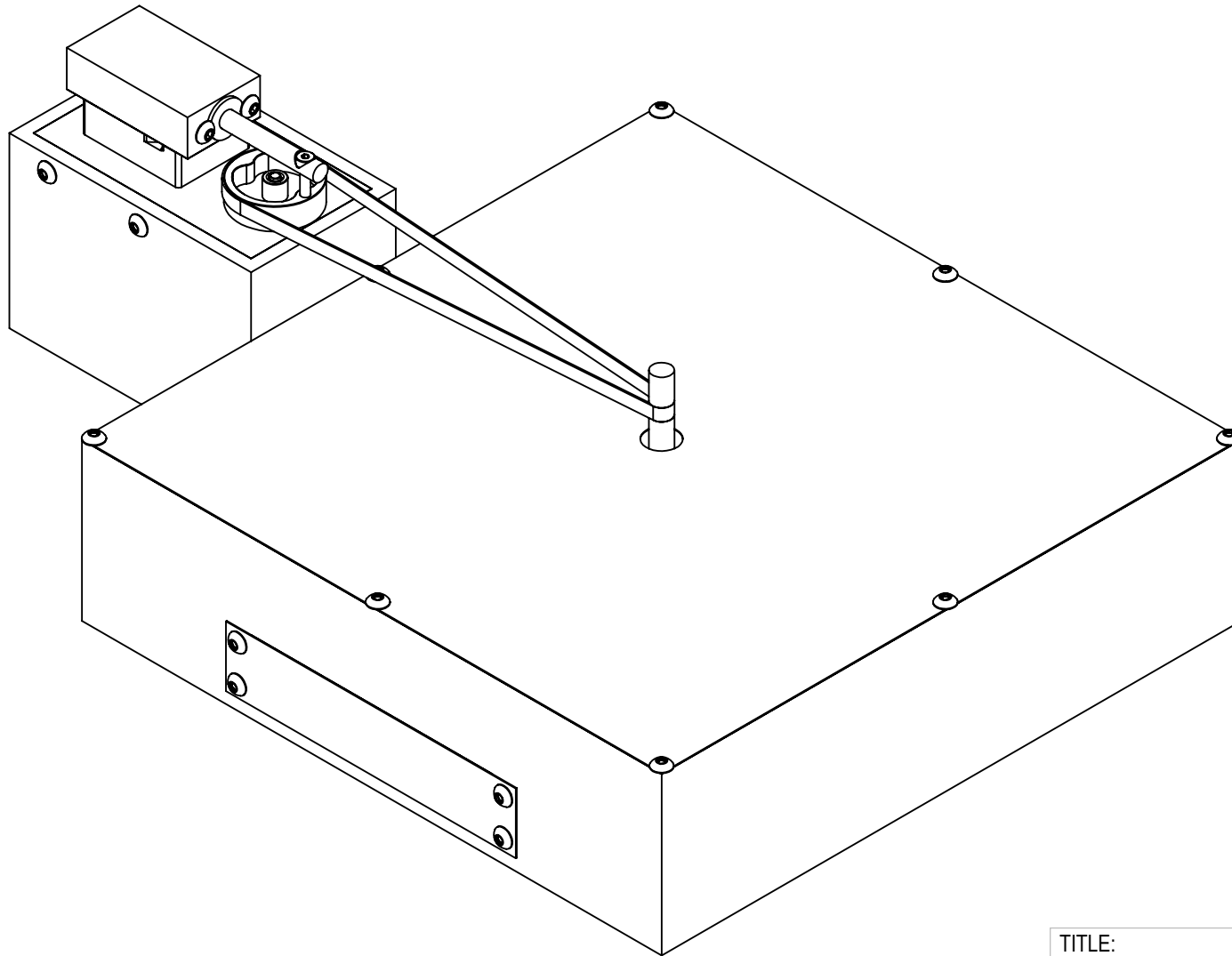
The food slicer and air engine are contained in a 3D printed base (P1) to prevent any components from moving around or interfering with the overall mechanism. The base was 3D printed because of its large size making it difficult to create with metal. Also, 3D printing the base allows for funkier design choices that would otherwise be difficult to manufacture. The base consists of 3 main parts: one part to hold the air engine and raise its height off the ground, another part to hold the slicer mechanism, and a third part to connect the previous two parts. In the air engine section, there are holes for four 10-24 screws (OTS4) made of 18-8 stainless steel that fasten to the 6061 aluminum air engine base, therefore restricting the ability of the air engine to move. In the slicer mechanism section, there is a hole at the bottom to allow for a steel ball bearing (OTS2) to press fit into the hole. This ball bearing is used to hold the shaft in place axially while still allowing it to rotate. The 3D printed base is made from PETG filament, which is food safe and more durable than standard PLA filament. Because food will often encounter the inner surface of the slicer mechanism section of the base, a food safe epoxy resin will be used to coat those surfaces, therefore ensuring that all surfaces that encounter food will not grow bacteria in the micropores of the base.

Inside of the food slicer section of the base, a plate (P4) is used for food to sit on. The plate has a hole in the middle to allow it to sit along the shaft, but a tolerance is included to ensure that the plate will not have any friction with the rotating shaft. One section of the plate will be cut out to allow sliced food to fall through into the bottom of the base to be collected after cutting. As the blades cut the food, a backing on the blade help push the food into this cut out portion of the plate. The plate is made of 6061 aluminum sheet metal because of its lower cost

and ease of machinability. To adjust the height of the plate for varying cut thicknesses, a set screw shaft collar made of black oxide 1215 carbon steel is used (OTS1). This shaft collar sits directly below the plate for support and slides up and down along the shaft. The user can slide the shaft collar to a desirable height to achieve a certain cut thickness, and an Allen wrench can be used to tighten the set screw and lock the cut thickness in place.

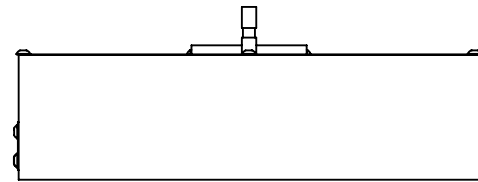
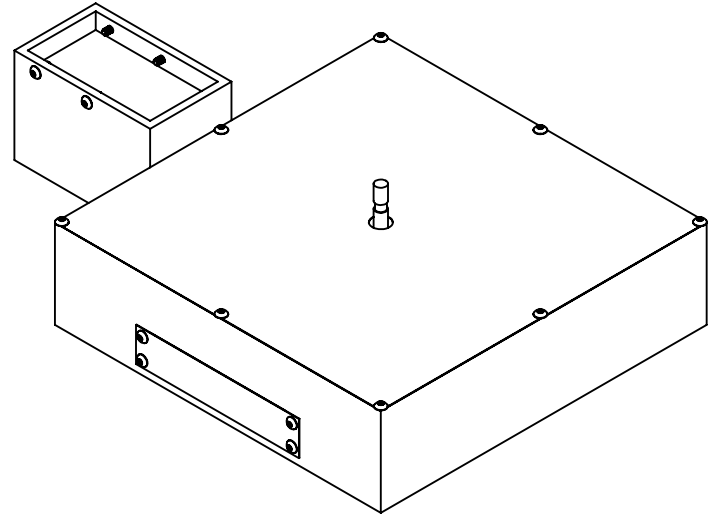
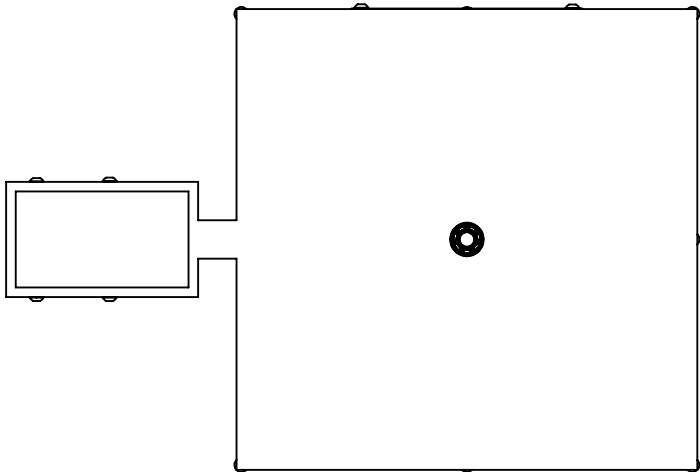
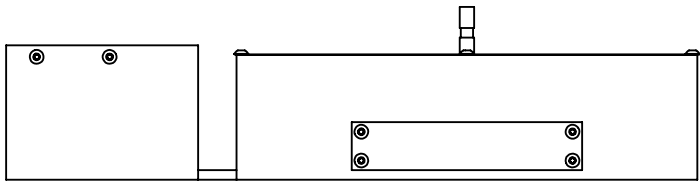
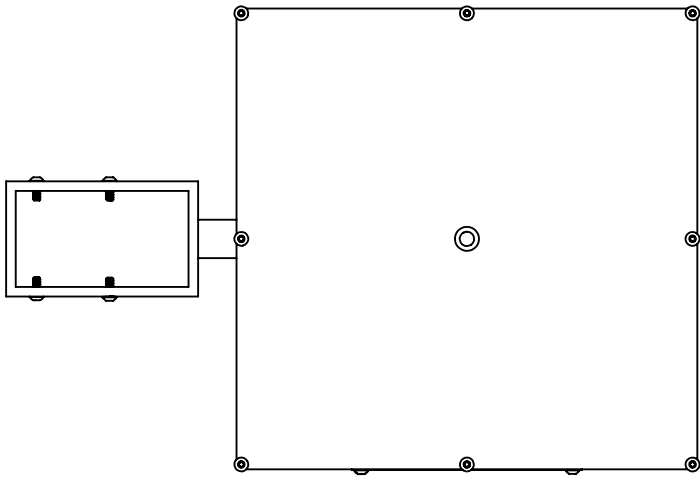
To make the design safe for consumers, a 6061 aluminum sheet metal safety cover (P2) is placed on top of the food slicer section of the base to ensure that the user cannot accidentally touch the sharp blades during operation. To keep the safety cover in place, eight 10-24 screws (OTS3) are used to fasten the safety cover into the base. A second safety feature is a hole on the outer surface of the base at the food slicer section. This hole will allow the user to adjust the shaft collar height without touching the blades. To cover this hole during operation, a 6061 aluminum sheet metal hole cover (P3) is used, and it is held in place with four 10-24 screws (3A).

Assembly Drawings and B.O.M.

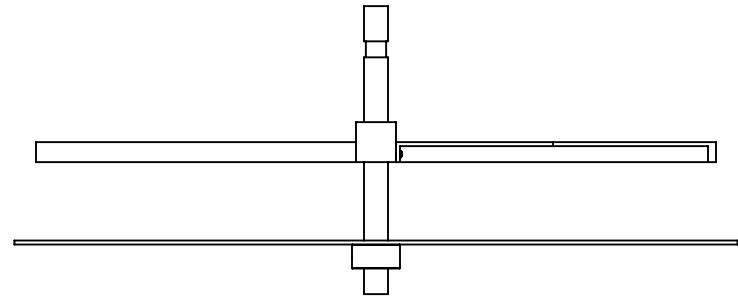
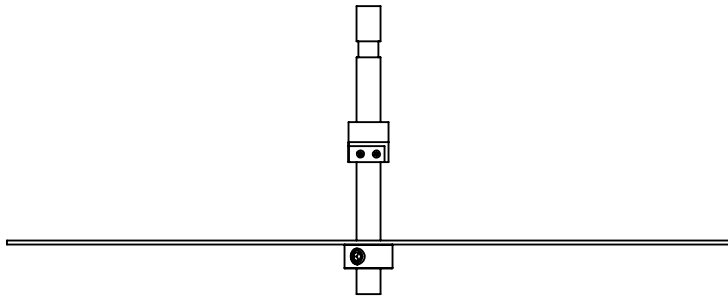
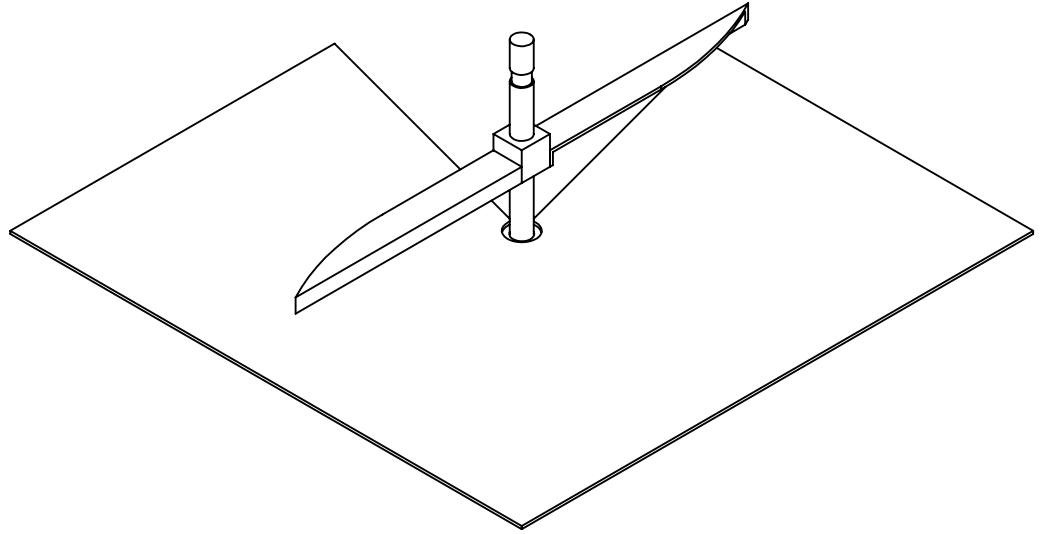
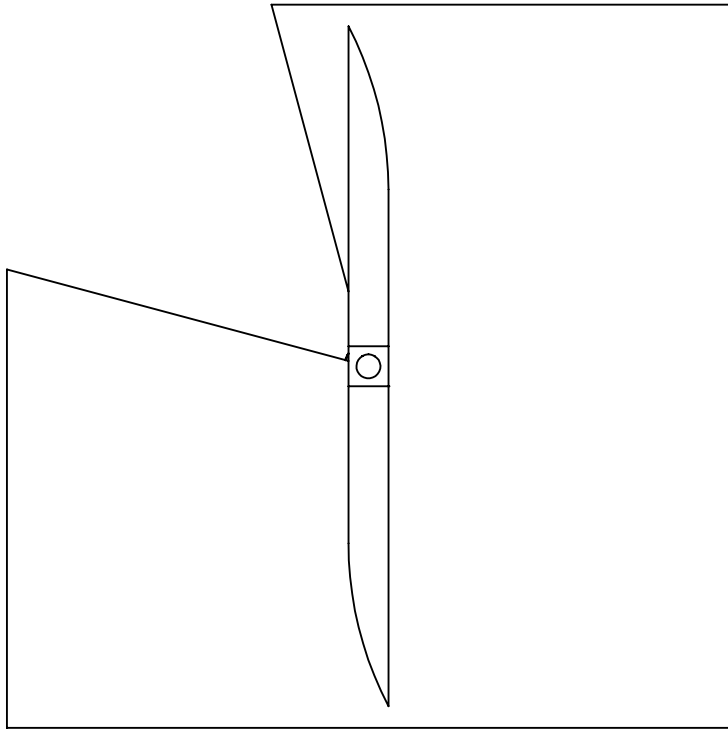


TITLE:				Slicer Assembly		
DRAWN				SYON AIDT		
CHECKED				JOSEPH VERMONT		
SIZE		DWG. NO.				REV
A		EML2322L-A1				A
SCALE: 2:5		SHEET 1 OF 5				

TOLERANCE UNLESS NOTED			
DIMENSION TYPE	PLACES IN DIMENSION		
	0.0	0.00	0.000
LOCATIONAL	±0.050	±0.020	±0.005
ANGULAR	±5	±2	±0.5

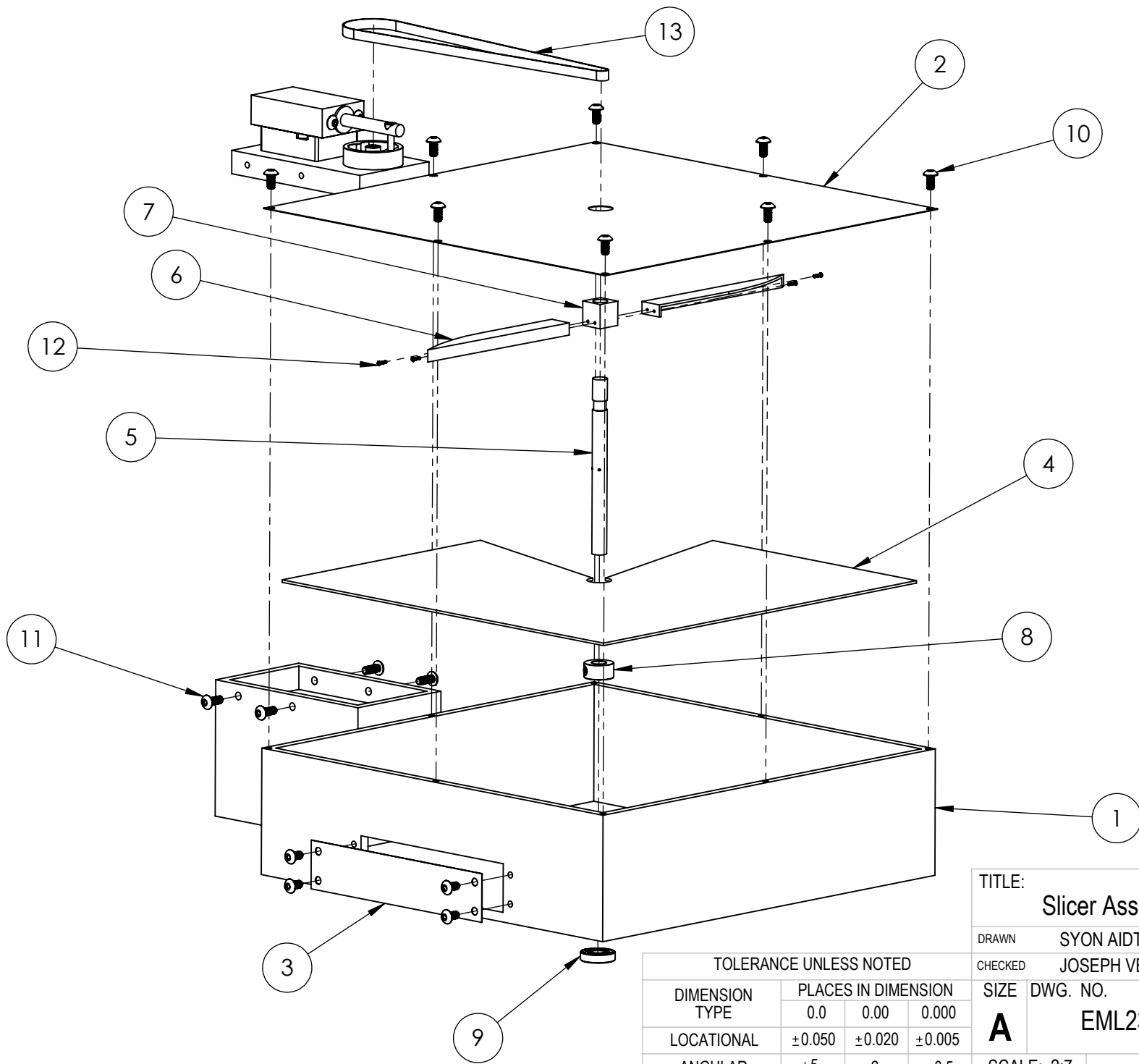


TITLE:				Slicer Concept Assembly		
DRAWN				SYON AIDT		
CHECKED				JOSEPH VERMONT		
SIZE		DWG. NO.				REV
A		EML2322L-A2				A
SCALE: 1:5		SHEET 2 OF 5				



TITLE:		Internal Subassembly		
DRAWN		SYON AIDT		
CHECKED		JOSEPH VERMONT		
SIZE	DWG. NO.	REV		
A	EML2322L-A3	A		
SCALE: 1:3		SHEET 3 OF 5		

TOLERANCE UNLESS NOTED			
DIMENSION TYPE	PLACES IN DIMENSION		
	0.0	0.00	0.000
LOCATIONAL	±0.050	±0.020	±0.005
ANGULAR	±5	±2	±0.5



TITLE: Slicer Assembly Exploded				
DRAWN SYON AIDT				
CHECKED JOSEPH VERMONT				
SIZE A	DWG. NO. EML2322L-A4			REV A
SCALE: 2:7		SHEET 4 OF 5		

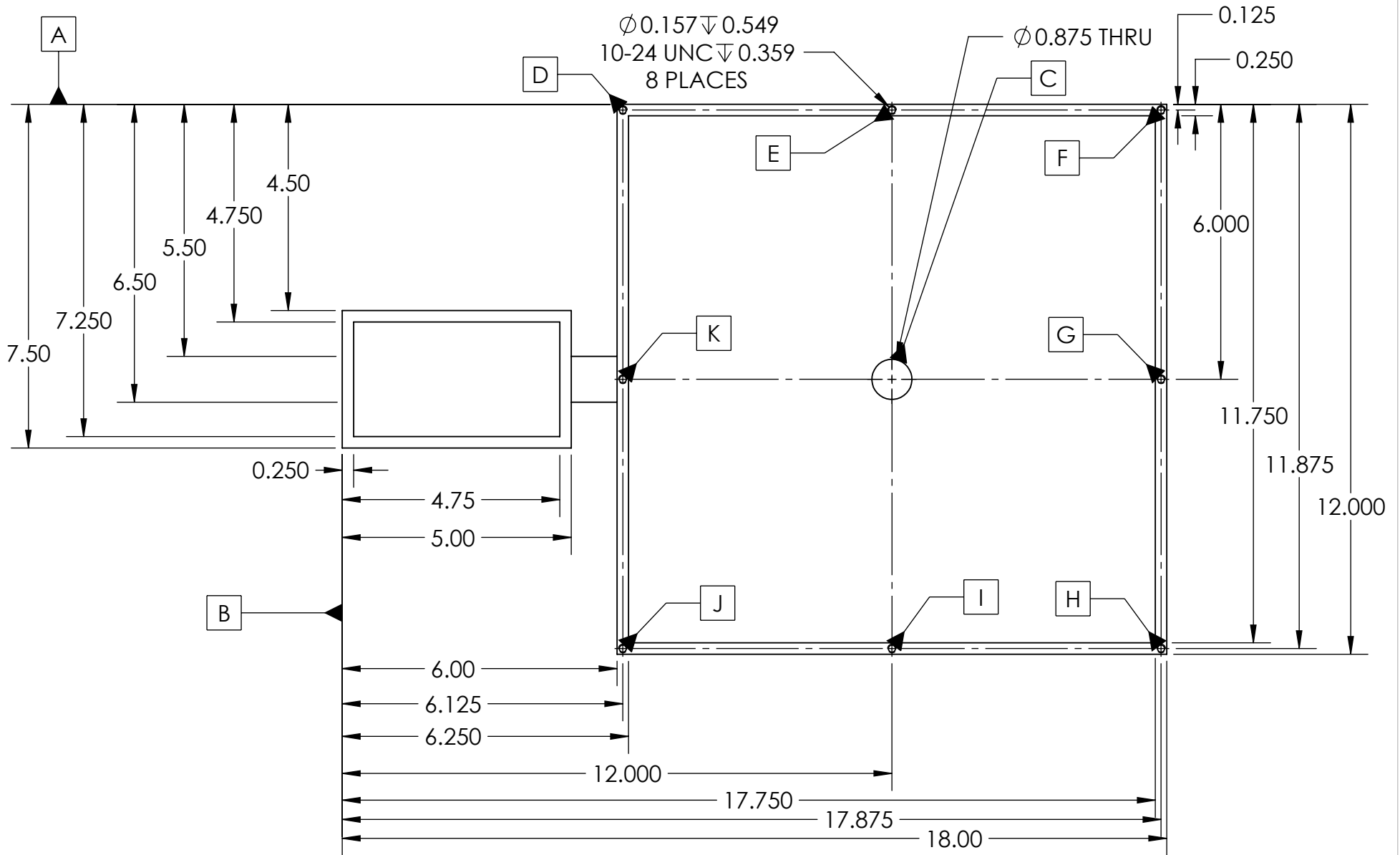
TOLERANCE UNLESS NOTED			
DIMENSION TYPE	PLACES IN DIMENSION		
	0.0	0.00	0.000
LOCATIONAL	±0.050	±0.020	±0.005
ANGULAR	±5	±2	±0.5

BILL OF MATERIALS				
ITEM NO.	PART NUMBER	VENDOR PART NUMBER	DESCRIPTION	QTY.
1	P1 (Base)	-	Base	1
2	P2 (Base Safety Cover)	-	Base Safety Cover	1
3	P3 (Base Hole Cover)	-	Base Hole Cover	1
4	P4 (Plate)	-	Plate	1
5	P5 (Shaft)	-	Shaft	1
6	P6 (Blade)	-	Blade	2
7	P7 (Blade Holder)	-	Blade Holder	1
8	OTS1	9414T8	3/8 inch ID x 3/4 inch OD x 3/8 inch Width Set Screw Shaft Collar	1
9	OTS2	60355K165	3/8 inch ID x 7/8 inch OD x 7/32 Width Open Ball Bearing	1
10	OTS3	92949A240	10-24 x 3/8 inch Hex Drive Button Head Screw	12
11	OTS4	92949A242	10-24 x 1/2 inch Hex Drive Button Head Screw	4
12	OTS5	92949A054	0-80 x 1/4 inch Hex Drive Button Head Screw	4
13	OTS6	6082K51	1/4 inch Width x 0.03 inch Thickness Wide Flat Belt	1

TITLE:				
Bill of Materials				
DRAWN		SYON AIDT		
CHECKED		JOSEPH VERMONT		
SIZE	DWG. NO.	REV		
A	EML2322L-A5	A		
SCALE: 1:1		SHEET 5 OF 5		

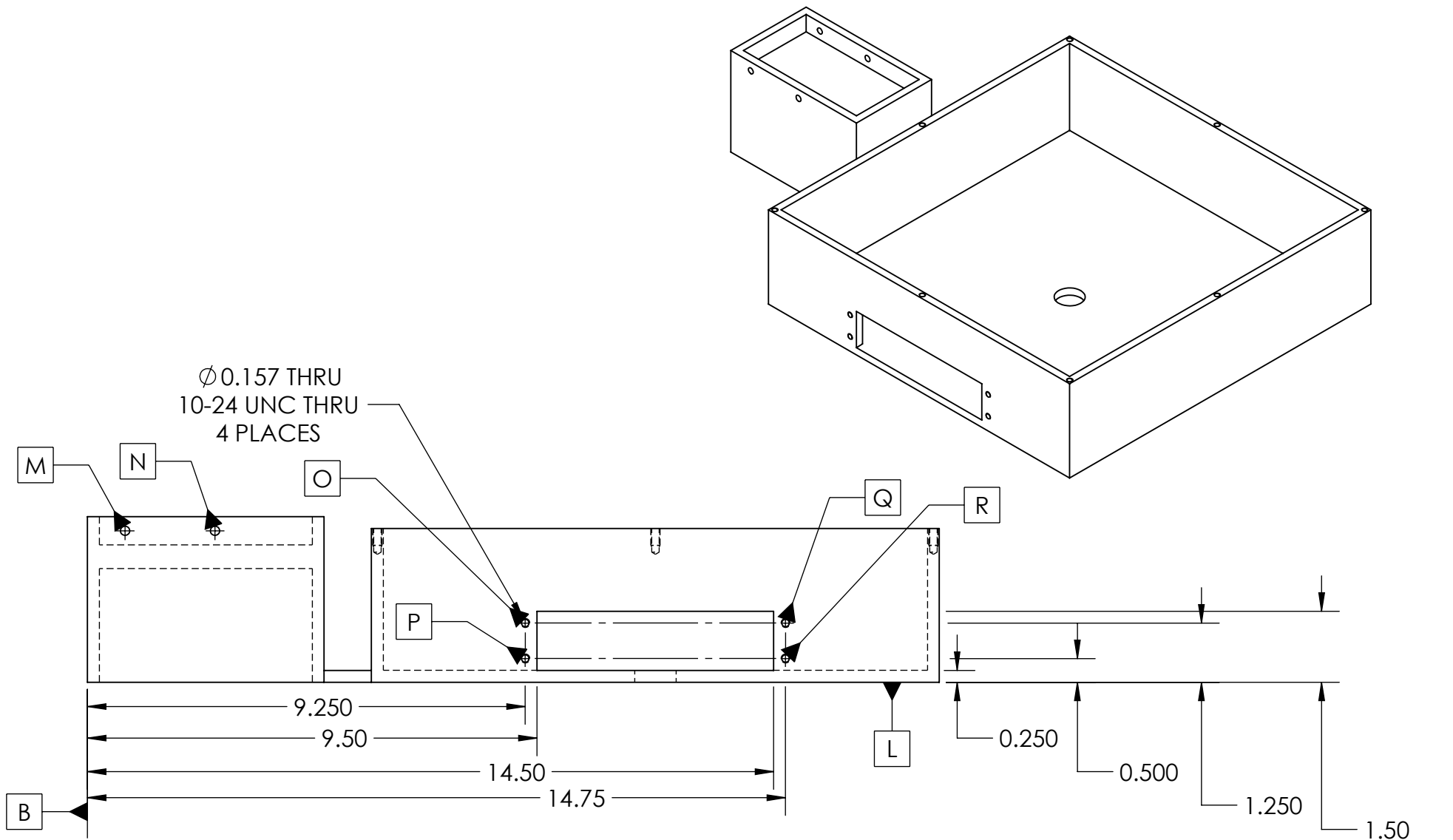
TOLERANCE UNLESS NOTED			
DIMENSION TYPE	PLACES IN DIMENSION		
	0.0	0.00	0.000
LOCATIONAL	±0.050	±0.020	±0.005
ANGULAR	±5	±2	±0.5

Appendix A: Detail Part Drawings

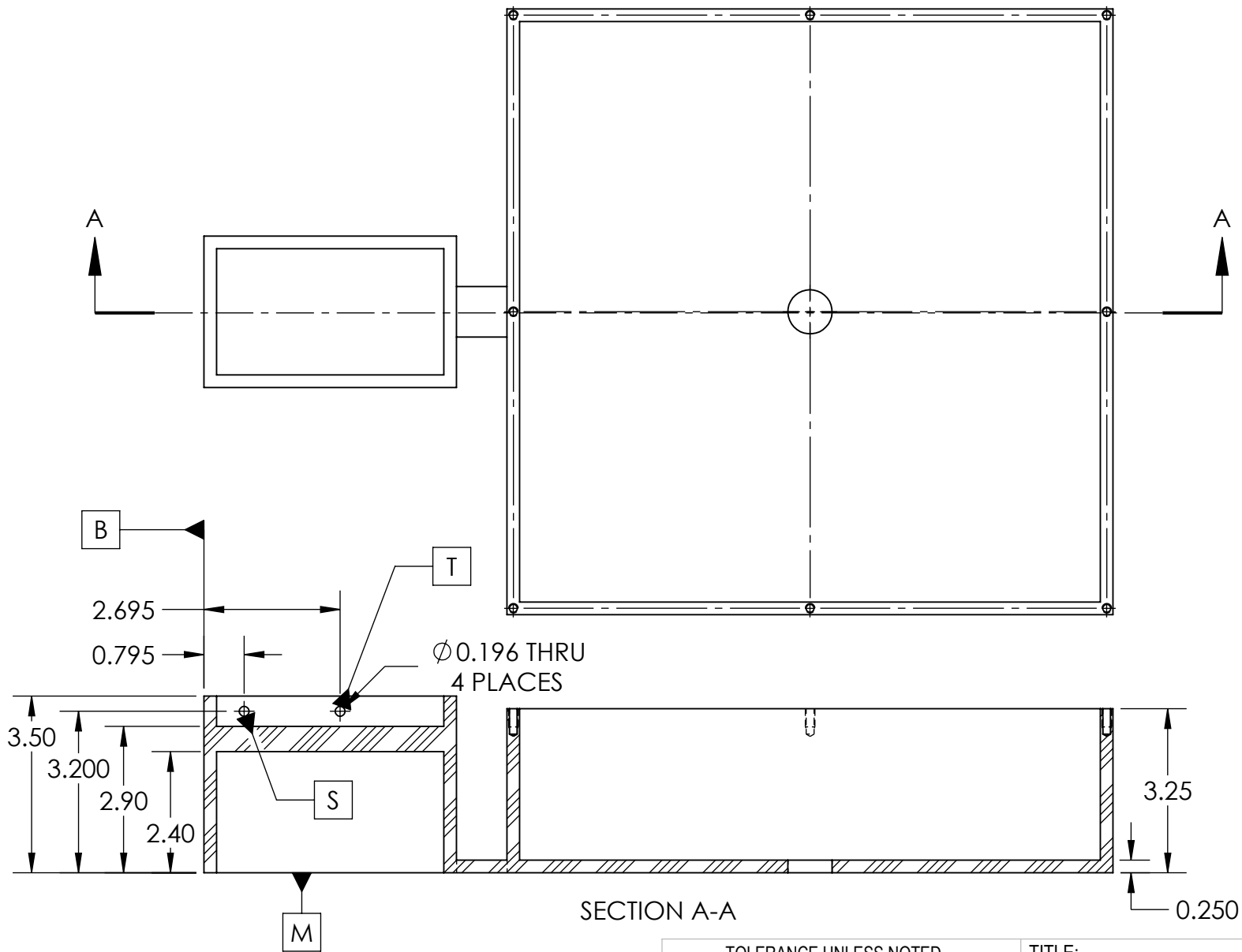


NOTES:
 1. QTY: 1
 2. DIMS IN INCHES
 3. MAT'L: PETG

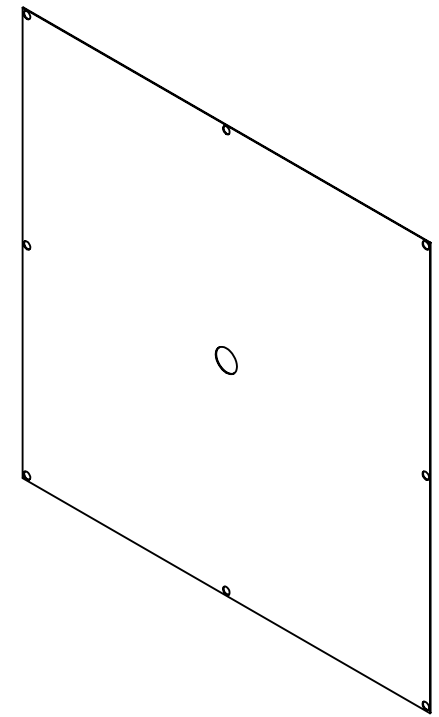
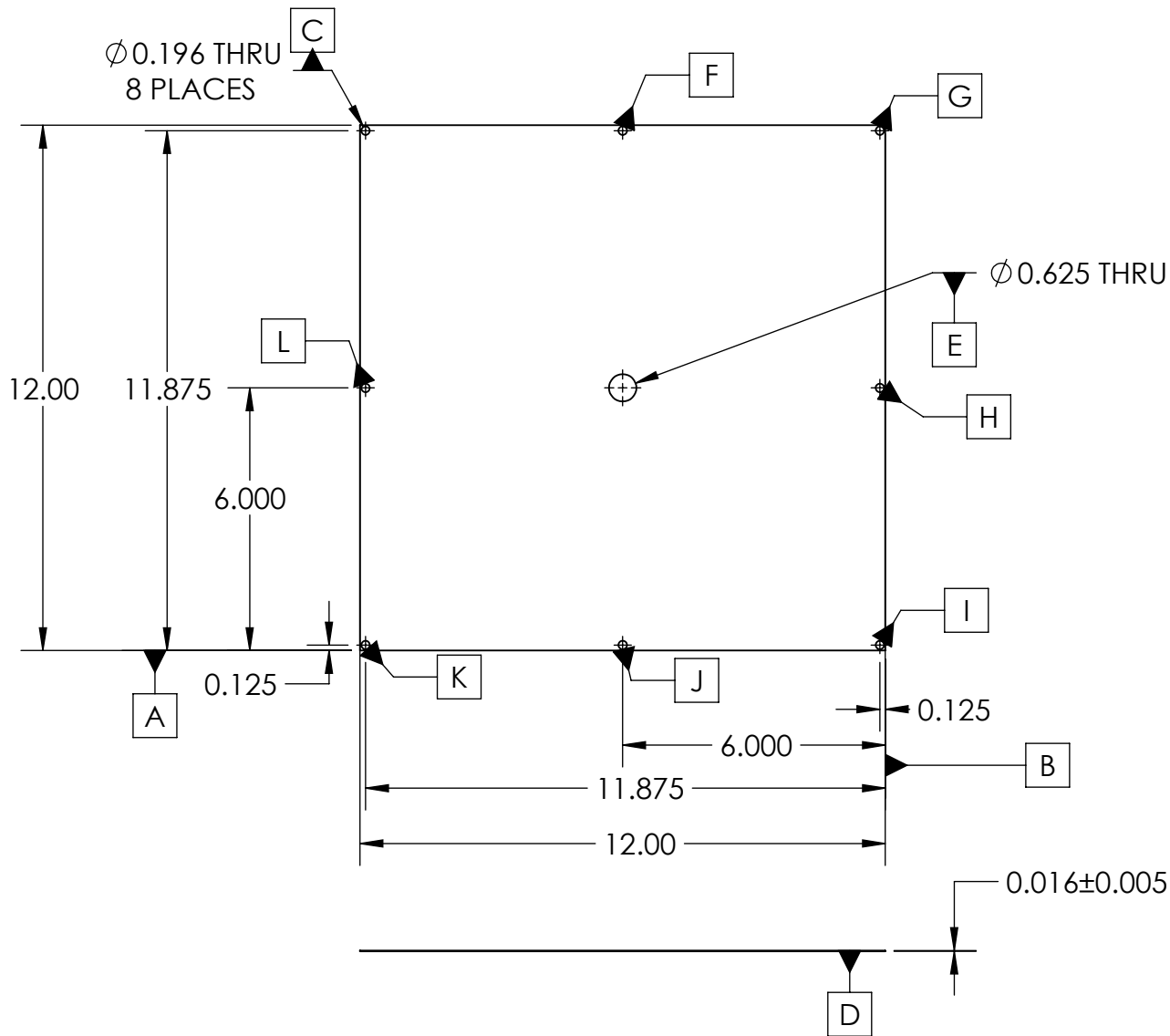
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OPERATION	PLACES IN DIMENSION			Base		
	0.0	0.00	0.000	DRAWN	SYON AIDT	
MACHINING	±0.050	±0.020	±0.005	CHECKED	ADAM GULDE	
CUT OFF (SAW, BURN, SHEAR)	±0.1	±0.060		SIZE	DWG. NO.	REV
WELDING	±0.1	±0.060		A	EML2322L-P1	A
ANGULAR DIMS	±5	±2	±0.5	SCALE: 1:3		SHEET 1 OF 3



TOLERANCE UNLESS NOTED				TITLE:		
OPERATION	PLACES IN DIMENSION			Base		
	0.0	0.00	0.000	DRAWN	SYON AIDT	
MACHINING	±0.050	±0.020	±0.005	CHECKED	ADAM GULDE	
CUT OFF (SAW, BURN, SHEAR)	±0.1	±0.060		SIZE	DWG. NO.	REV
WELDING	±0.1	±0.060		A	EML2322L-P1	A
ANGULAR DIMS	±5	±2	±0.5	SCALE: 1:3	SHEET 2 OF 3	

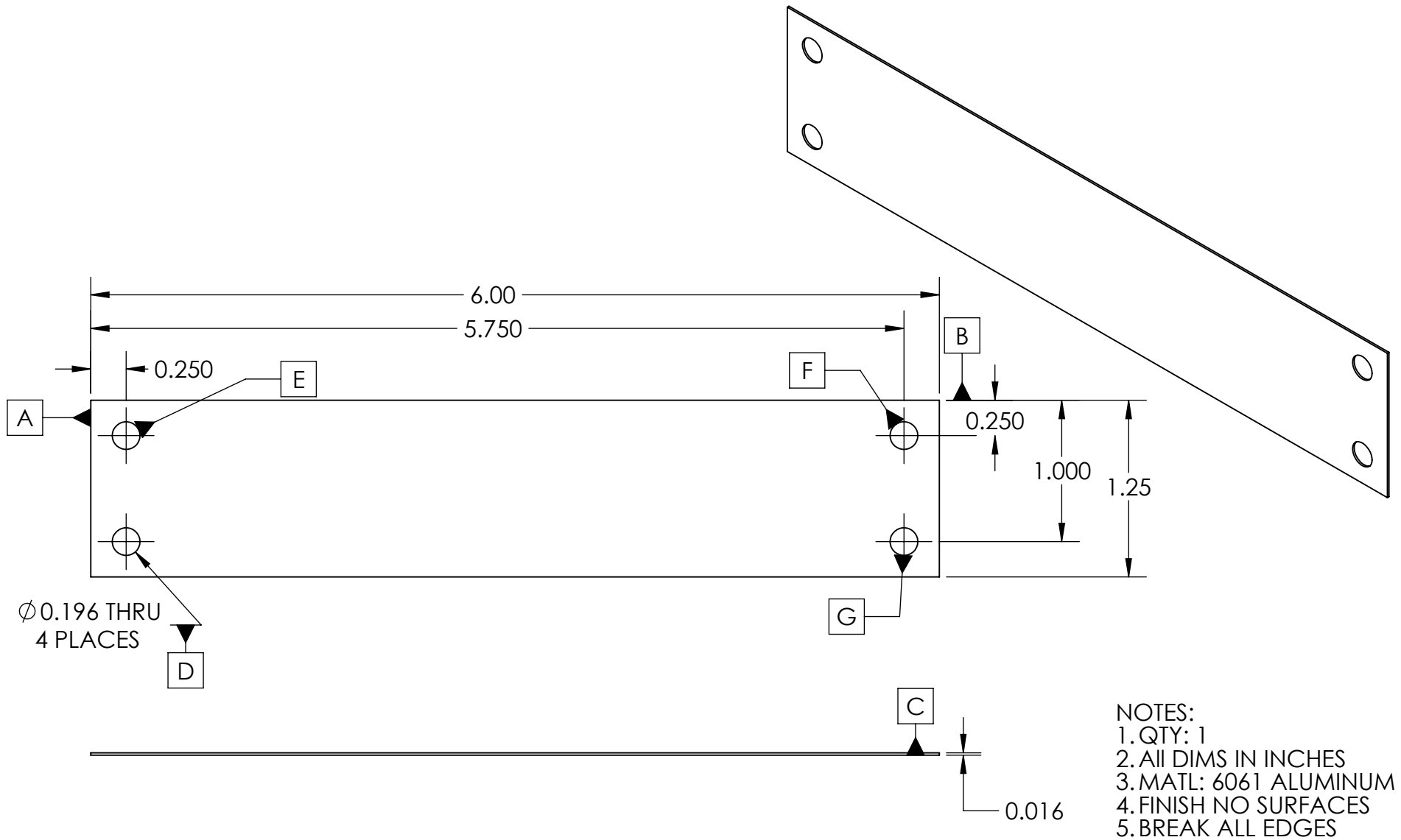


TOLERANCE UNLESS NOTED				TITLE: Base		
OPERATION	PLACES IN DIMENSION			DRAWN	SYON AIDT	
	0.0	0.00	0.000	CHECKED	ADAM GULDE	
MACHINING	±0.050	±0.020	±0.005	SIZE	DWG. NO.	REV
CUT OFF (SAW, BURN, SHEAR)	±0.1	±0.060		A	EML2322L-P1	A
WELDING	±0.1	±0.060		SCALE: 1:3		
ANGULAR DIMS	±5	±2	±0.5			SHEET 3 OF 3

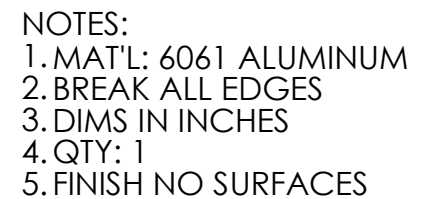


- NOTES:
- 1.QTY: 1
 - 2.MAT'L: 6061 ALUMINUM
 - 3.FINISH NO SURFACES
 - 4.BREAK ALL EDGES
 - 5.DIMS IN INCHES

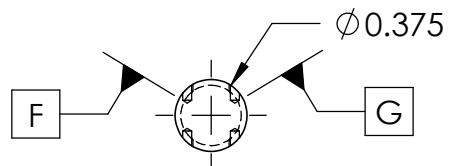
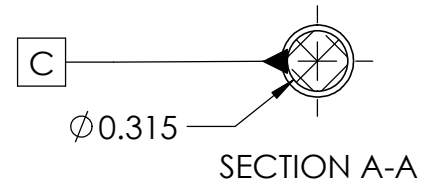
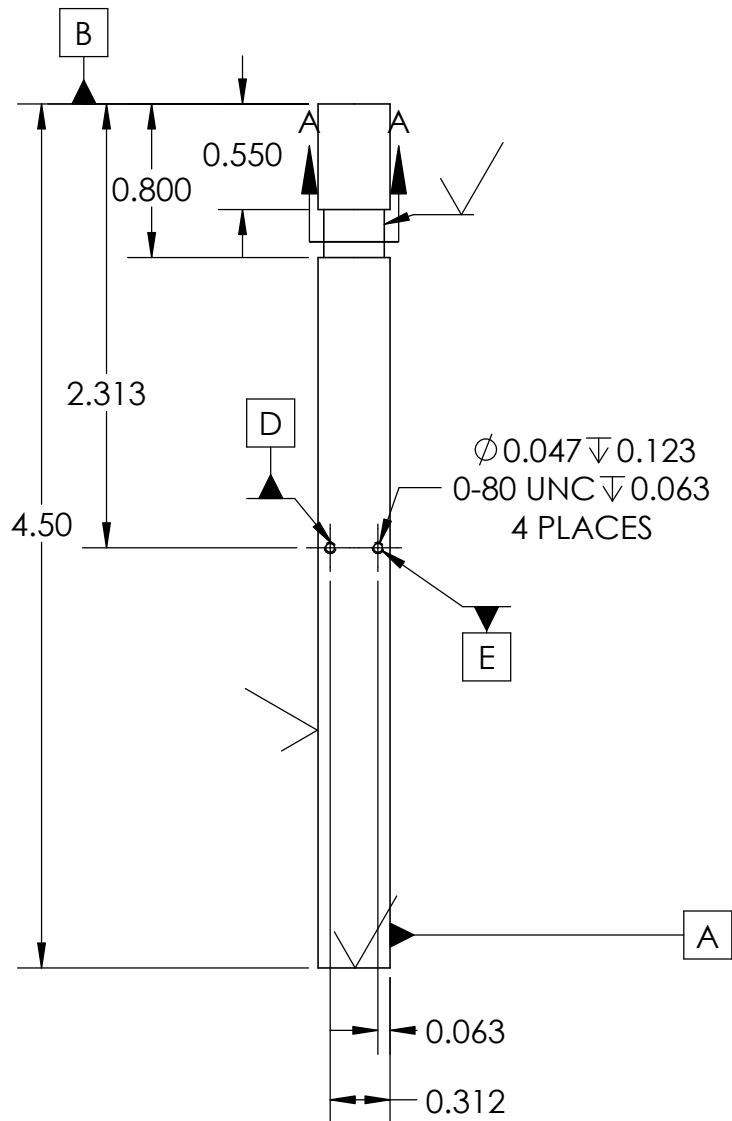
TOLERANCE UNLESS NOTED				TITLE:		
OPERATION	PLACES IN DIMENSION			Base Safety Cover		
	0.0	0.00	0.000	DRAWN	MILAN STEVANOVIC	
MACHINING	±0.050	±0.020	±0.005	CHECKED	SYON AIDT	
CUT OFF (SAW, BURN, SHEAR)	±0.1	±0.060		SIZE	DWG. NO.	REV
WELDING	±0.1	±0.060		A	EML2322L-P2	A
ANGULAR DIMS	±5	±2	±0.5	SCALE: 1:4	SHEET 1 OF 1	



TOLERANCE UNLESS NOTED				TITLE:		
OPERATION	PLACES IN DIMENSION			Base Hole Cover		
	0.0	0.00	0.000	DRAWN	MILAN STEVANOVIC	
MACHINING	±0.050	±0.020	±0.005	CHECKED	SYON AIDT	
CUT OFF (SAW, BURN, SHEAR)	±0.1	±0.060		SIZE	DWG. NO.	REV
WELDING	±0.1	±0.060		A	EML2322L-P3	A
ANGULAR DIMS	±5	±2	±0.5	SCALE: 1:1		SHEET 1 OF 1

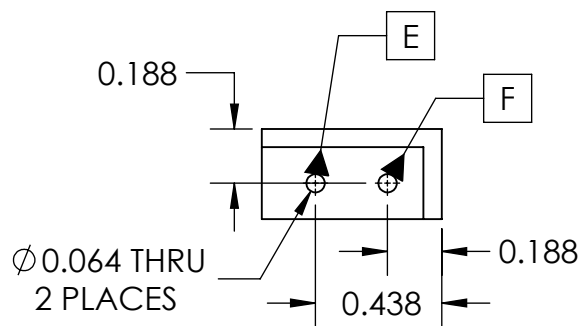
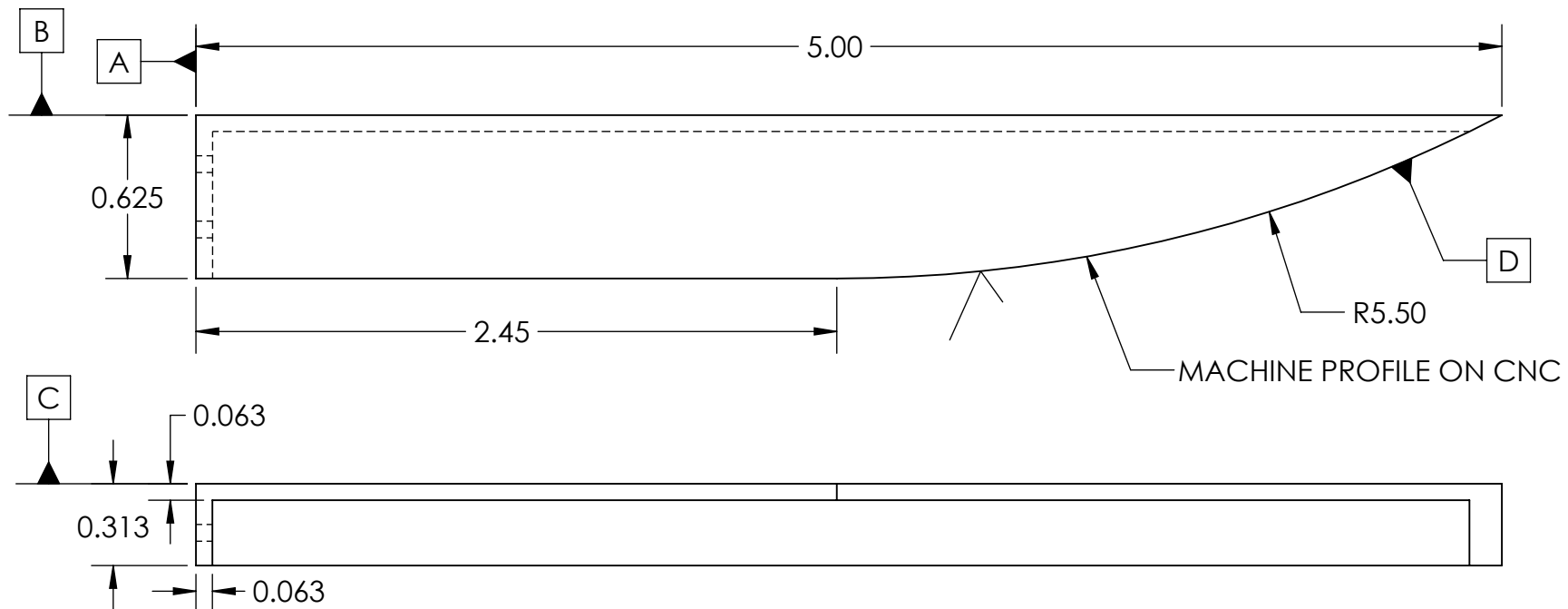


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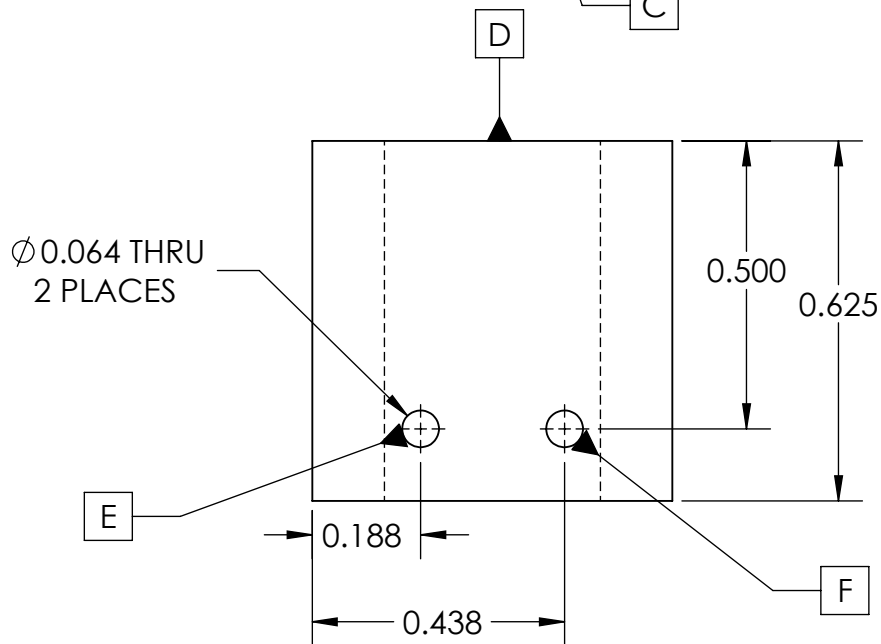
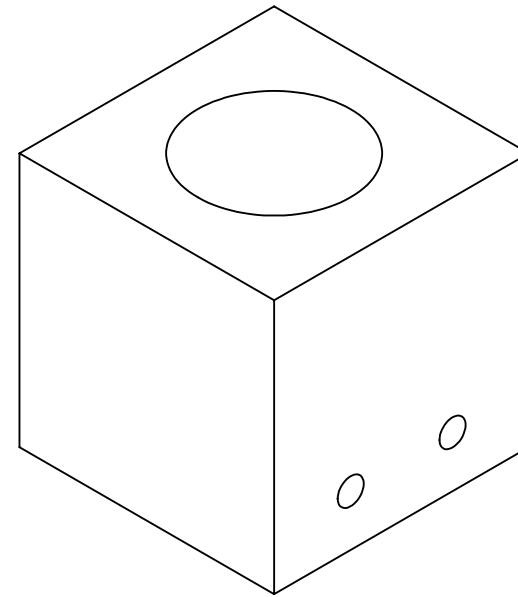
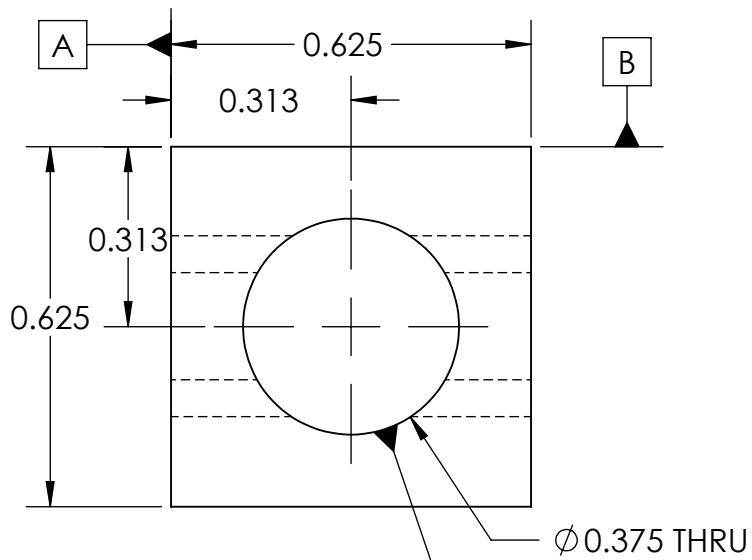
- NOTES:
1. DIMS IN INCHES
 2. FINISH NOTED SURFACES
 3. QTY: 1
 4. MAT'L: 6061 ALUMINUM
 5. BREAK ALL EDGES

TOLERANCE UNLESS NOTED				TITLE: Shaft		
OPERATION	PLACES IN DIMENSION			DRAWN	ADAM GULDE	
	0.0	0.00	0.000	CHECKED	MILAN STEVANOVIC	
MACHINING	± 0.050	± 0.020	± 0.005	SIZE	DWG. NO.	REV
CUT OFF (SAW, BURN, SHEAR)	± 0.1	± 0.060		A	EML2322L-P5	A
WELDING	± 0.1	± 0.060		SCALE: 1:1		
ANGULAR DIMS	± 5	± 2	± 0.5			SHEET 1 OF 1



NOTES:
 1. QTY: 2
 2. DIMS IN INCHES
 3. MAT'L: 304 STAINLESS STEEL
 4. BREAK ALL EDGES
 5. FINISH NOTED SURFACES

TOLERANCE UNLESS NOTED				TITLE:		
OPERATION	PLACES IN DIMENSION			Blade		
	0.0	0.00	0.000	DRAWN	Hayden Patalinghug	
MACHINING	± 0.050	± 0.020	± 0.005	CHECKED	Syon Aidt	
CUT OFF (SAW, BURN, SHEAR)	± 0.1	± 0.060		SIZE	DWG. NO.	REV
WELDING	± 0.1	± 0.060		A	EML2322L-P6	A
ANGULAR DIMS	± 5	± 2	± 0.5	SCALE: 3:2	SHEET 1 OF 1	



- NOTES:
1. QTY: 1
 2. DIMS IN INCHES
 3. MAT'L: 6061 ALUMINUM
 4. FINISH NO SURFACES
 5. BREAK ALL EDGES

TOLERANCE UNLESS NOTED				TITLE: Blade Holder		
OPERATION	PLACES IN DIMENSION			DRAWN Hayden Patalinghug		
	0.0	0.00	0.000	CHECKED Syon Aidt		
MACHINING	±0.050	±0.020	±0.005	SIZE DWG. NO. EML2322L-P7		
CUT OFF (SAW, BURN, SHEAR)	±0.1	±0.060		REV A		
WELDING	±0.1	±0.060				
ANGULAR DIMS	±5	±2	±0.5	SCALE: 3:1		

Appendix B: Manufacturing Inspection Records

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P1		A	Base		7C				
							Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
1	A	Linear Dimension	Caliper	4.50	4.48	4.52	in		
2	A	Linear Dimension	Caliper	4.750	4.745	4.755	in		
3	A	Linear Dimension	Caliper	5.50	5.48	5.52	in		
4	A	Linear Dimension	Caliper	6.50	6.48	6.52	in		
5	A	Linear Dimension	Caliper	7.250	7.245	7.255	in		
6	A	Linear Dimension	Caliper	7.50	7.48	7.52	in		
7	A	Linear Dimension	Caliper	0.125	0.120	0.130	in		
8	A	Linear Dimension	Caliper	0.250	0.245	0.255	in		
9	A	Linear Dimension	Caliper	6.000	5.995	6.005	in		
10	A	Linear Dimension	Caliper	11.750	11.745	11.755	in		
11	A	Linear Dimension	Caliper	11.875	11.870	11.880	in		
12	A	Linear Dimension	Caliper	12.000	11.995	12.005	in		
13	B	Linear Dimension	Caliper	0.250	0.245	0.255	in		
14	B	Linear Dimension	Caliper	4.75	4.73	4.77	in		
15	B	Linear Dimension	Caliper	5.00	4.98	5.02	in		
RESULT:							PASS		
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3		Hayden Patalinghug		TA Signature					
Group Member 4		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P1		A	Base		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
16	B	Linear Dimension	Caliper	6.00	5.98	6.02	in		
17	B	Linear Dimension	Caliper	6.125	6.120	6.130	in		
18	B	Linear Dimension	Caliper	6.250	6.245	6.255	in		
19	B	Linear Dimension	Caliper	12.000	11.995	12.005	in		
20	B	Linear Dimension	Caliper	17.750	11.745	11.755	in		
21	B	Linear Dimension	Caliper	17.875	17.870	11.880	in		
22	B	Linear Dimension	Caliper	18.00	17.98	18.02	in		
23	C	Diameter	Caliper	0.875	0.870	0.880	in		
24	C	Depth	Visual	THRU	-	-	-		
25	D	Diameter	Caliper	0.157	0.152	0.162	in		
26	D	Depth	Caliper	0.549	0.544	0.554	in		
27	D	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
28	D	Depth	Caliper	0.359	0.354	0.364	in		
29	E	Diameter	Caliper	0.157	0.152	0.162	in		
30	E	Depth	Caliper	0.549	0.544	0.554	in		
RESULT:							PASS		
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3		Hayden Patalinghug		TA Signature					
Group Member 4		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P1		A	Base		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
31	E	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
32	E	Depth	Caliper	0.359	0.354	0.364	in		
33	F	Diameter	Caliper	0.157	0.152	0.162	in		
34	F	Depth	Caliper	0.549	0.544	0.554	in		
35	F	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
36	F	Depth	Caliper	0.359	0.354	0.364	in		
37	G	Diameter	Caliper	0.157	0.152	0.162	in		
38	G	Depth	Caliper	0.549	0.544	0.554	in		
39	G	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
40	G	Depth	Caliper	0.359	0.354	0.364	in		
41	H	Diameter	Caliper	0.157	0.152	0.162	in		
42	H	Depth	Caliper	0.549	0.544	0.554	in		
43	H	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
44	H	Depth	Caliper	0.359	0.354	0.364	in		
45	I	Diameter	Caliper	0.157	0.152	0.162	in		
RESULT:							PASS		
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3:		Hayden Patalinghug		TA Signature					
Group Member 4:		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P1		A	Base		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
46	I	Depth	Caliper	0.549	0.544	0.554	in		
47	I	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
48	I	Depth	Caliper	0.359	0.354	0.364	in		
49	J	Diameter	Caliper	0.157	0.152	0.162	in		
50	J	Depth	Caliper	0.549	0.544	0.554	in		
51	J	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
52	J	Depth	Caliper	0.359	0.354	0.364	in		
53	K	Diameter	Caliper	0.157	0.152	0.162	in		
54	K	Depth	Caliper	0.549	0.544	0.554	in		
55	K	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
56	K	Depth	Caliper	0.359	0.354	0.364	in		
57	B	Linear Dimension	Caliper	9.250	9.245	9.255	in		
58	B	Linear Dimension	Caliper	9.50	9.48	9.52	in		
59	B	Linear Dimension	Caliper	14.50	14.48	14.52	in		
60	B	Linear Dimension	Caliper	14.75	14.73	14.77	in		
RESULT:							PASS		
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3:		Hayden Patalinghug		TA Signature					
Group Member 4:		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P1		A	Base		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
61	L	Linear Dimension	Caliper	0.250	0.245	0.255	in		
62	L	Linear Dimension	Caliper	0.500	0.495	0.505	in		
63	L	Linear Dimension	Caliper	1.250	1.245	1.255	in		
64	L	Linear Dimension	Caliper	1.50	1.48	1.52	in		
65	M	Diameter	Caliper	0.196	0.191	0.201	in		
66	M	Depth	Visual	THRU	-	-	-		
67	N	Diameter	Caliper	0.196	0.191	0.201	in		
68	N	Depth	Visual	THRU	-	-	-		
69	O	Diameter	Caliper	0.157	0.152	0.162	in		
70	O	Depth	Visual	THRU	-	-	-		
71	O	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
72	O	Depth	Visual	THRU	-	-	-		
73	P	Diameter	Caliper	0.157	0.152	0.162	in		
74	P	Depth	Visual	THRU	-	-	-		
75	P	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
RESULT:								PASS	
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3:		Hayden Patalinghug							
Group Member 4:		Milan Stevanovic		TA Signature					

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P1		A	Base		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
76	P	Depth	Visual	THRU	-	-	in		
77	Q	Diameter	Caliper	0.157	0.152	0.162	in		
78	Q	Depth	Visual	THRU	-	-	-		
79	Q	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
80	Q	Depth	Visual	THRU	-	-	-		
81	R	Diameter	Caliper	0.157	0.152	0.162	in		
82	R	Depth	Visual	THRU	-	-	-		
83	R	Thread	Thread Gauge or Fastener	10-24 UNC	-	-	-		
84	R	Depth	Visual	THRU	-	-	-		
85	B	Linear Dimension	Caliper	2.695	2.690	2.700	in		
86	B	Linear Dimension	Caliper	0.795	0.790	0.800	in		
87	M	Linear Dimension	Caliper	2.40	2.38	2.42	in		
88	M	Linear Dimension	Caliper	2.90	2.88	2.92	in		
89	M	Linear Dimension	Caliper	3.200	3.195	3.205	in		
90	M	Linear Dimension	Caliper	3.50	3.48	3.52	in		
RESULT:								PASS	
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3		Hayden Patalinghug		TA Signature					
Group Member 4		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P1		A	Base		7C				
							Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
91	M	Linear Dimension	Caliper	0.250	0.245	0.255	in		
92	M	Linear Dimension	Caliper	3.25	3.23	3.27	in		
93	S	Diameter	Caliper	0.196	0.191	0.201	in		
94	S	Depth	Visual	THRU	-	-	-		
95	T	Diameter	Caliper	0.196	0.191	0.201	in		
96	T	Depth	Visual	THRU	-	-	-		
RESULT:								PASS	
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3		Hayden Patalinghug			TA Signature				
Group Member 4		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L-P2		A	Base Safety Cover		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
1	A	linear dimension	caliper	0.125	0.120	0.130	inches		
2	A	linear dimension	caliper	6.000	5.995	6.005	inches		
3	A	linear dimension	caliper	11.875	11.870	11.880	inches		
4	A	linear dimension	caliper	12.00	11.94	12.06	inches		
5	B	linear dimension	caliper	0.125	0.120	0.130	inches		
6	B	linear dimension	caliper	6.000	5.995	6.005	inches		
7	B	linear dimension	caliper	11.875	11.870	11.880	inches		
8	B	linear dimension	caliper	12.00	11.94	12.06	inches		
9	C	diameter	caliper	0.196	0.191	0.201	inches		
10	D	linear dimension	caliper	0.016	0.011	0.021	inches		
11	E	diameter	caliper	0.625	0.620	0.630	inches		
12	F	diameter	caliper	0.196	0.191	0.201	inches		
13	G	diameter	caliper	0.196	0.191	0.201	inches		
14	H	diameter	caliper	0.196	0.191	0.201	inches		
15	I	diameter	caliper	0.196	0.191	0.201	inches		
16	J	diameter	caliper	0.196	0.191	0.201	inches		
17	K	diameter	caliper	0.196	0.191	0.201	inches		
18	L	diameter	caliper	0.196	0.191	0.201	inches		
19	Note	note	visual	break all edges	-	-			
20	Note	note	visual	finish no surfaces	-	-			
RESULT:								PASS	
Inspector Signature:					FAIL	☐ Remanufacture			
Group Member 1:						☐ Rework			
Group Member 2:						☐ Engineering Change			
Group Member 3		Hayden Patalinghug		TA Signature					
Group Member 4		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L - P3		A	Base Hole Cover		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
1	A	Linear Dimension	caliper	6.00	5.94	6.06	inches		
2	A	Linear Dimension	caliper	5.750	5.745	5.755	inches		
3	A	Linear Dimension	caliper	0.250	0.245	0.255	inches		
4	B	Linear Dimension	caliper	1.25	1.19	1.31	inches		
5	B	Linear Dimension	caliper	1.000	0.995	1.005	inches		
5	B	Linear Dimension	caliper	0.250	0.245	0.255	inches		
6	C	Linear Dimension	caliper	0.016	0.011	0.021	inches		
7	D	Diameter	caliper	0.196	0.191	0.201	inches		
8	E	Diameter	caliper	0.196	0.191	0.201	inches		
9	F	Diameter	caliper	0.196	0.191	0.201	inches		
10	G	Diameter	caliper	0.196	0.191	0.201	inches		
11	Note	note	visual	break all edges	-	-			
12	Note	note	visual	finish no surfaces	-	-			
RESULT:								PASS	
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Syon Aidt						☐ Rework	
Group Member 2:		Adam Gulde						☐ Engineering Change	
Group Member 3		Hayden Patalinghug		TA Signature					
Group Member 4		Milan Stevanovic							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L-P4		A	Plate		7C				
							Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
1	A	Linear Dimension	Caliper	11.3	11.2	11.4	inch		
2	A	Linear Dimension	Caliper	5.650	5.645	5.655	inch		
3	A	Linear Dimension	Caliper	7.16	7.10	7.22	inch		
4	B	Linear Dimension	Caliper	11.3	11.2	11.4	inch		
5	B	Linear Dimension	Caliper	5.650	5.645	5.655	inch		
6	B	Linear Dimension	Caliper	7.16	7.10	7.22	inch		
7	C	Hole Diameter	Caliper	0.625	0.620	0.630	inch		
8	C	Hole Depth	Visual	THRU	-	-	-		
9	D	Angle	Digital Angle Finder	60.0	55.0	65.0	degrees		
10	E	Linear Dimension	Caliper	0.063	0.058	0.068	inch		
11	Note	Note	Visual	Break all edges	-	-	-		
12	Note	Note	Visual	Finish no surfaces	-	-	-		
RESULT:								PASS	
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Adam Gulde						☐ Rework	
Group Member 2:		Hayden Patalinghug						☐ Engineering Change	
Group Member 3		Milan Stevanovic		TA Signature					
Group Member 4		Syon Aidt							

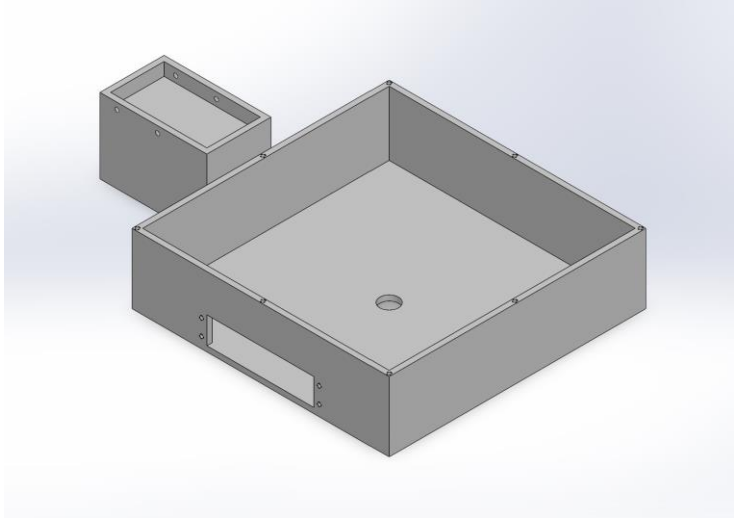
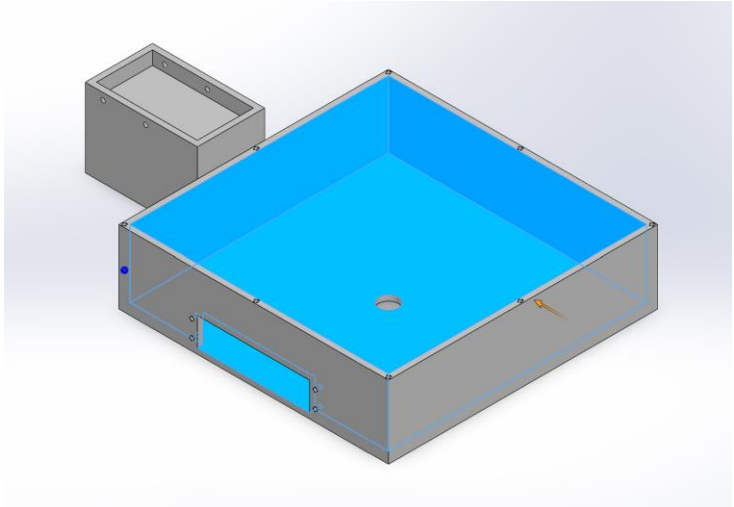
EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML2322L-P5		A	Shaft		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
1	A	Linear Dimension	Caliper	0.375	0.370	0.380	inch		
2	A	Linear Dimension	Caliper	0.063	0.058	0.068	inch		
3	A	Linear Dimension	Caliper	0.312	0.307	0.317	inch		
4	B	Linear Dimension	Caliper	0.550	0.545	0.555	inch		
5	B	Linear Dimension	Caliper	0.800	0.795	0.805	inch		
6	B	Linear Dimension	Caliper	2.313	2.308	2.318	inch		
7	B	Linear Dimension	Caliper	4.50	4.48	4.52	inch		
12	C	Linear Dimension	Caliper	0.315	0.310	0.320	inch		
13	D	Hole Diameter	Caliper	0.047	0.042	0.052	inch		
14	D	Hole Depth	Caliper	0.123	0.118	0.128	inch		
15	E	Hole Diameter	Caliper	0.047	0.042	0.052	inch		
16	E	Hole Depth	Caliper	0.123	0.118	0.128	inch		
17	F	Hole Diameter	Caliper	0.047	0.042	0.052	inch		
18	F	Hole Depth	Caliper	0.123	0.118	0.128	inch		
19	G	Hole Diameter	Caliper	0.047	0.042	0.052	inch		
20	G	Hole Depth	Caliper	0.123	0.118	0.128	inch		
21	D	Threads	Fastener OR Thread Gauge	0-80 UNC	-	-	-		
22	E	Threads	Fastener OR Thread Gauge	0-80 UNC	-	-	-		
23	F	Threads	Fastener OR Thread Gauge	0-80 UNC	-	-	-		
24	G	Threads	Fastener OR Thread Gauge	0-80 UNC	-	-	-		
25	Note	Note	Visual	Break all edges	-	-	-		
26	Note	Note	Visual	Finish no surfaces	-	-	-		
RESULT:								PASS	
Inspector Signature:						FAIL		☐ Remanufacture	
Group Member 1:		Adam Gulde						☐ Rework	
Group Member 2:		Hayden Patalinghug						☐ Engineering Change	
Group Member 3		Milan Stevanovic		TA Signature					
Group Member 4		Syon Aidt							

EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML 2322L-P6		A	Blade		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
1	A	Linear Dimension	Caliper	5.00	4.98	5.02	in		
2	A	Linear Dimension	Caliper	2.45	2.43	2.47	in		
3	A	Linear Dimension	Caliper	0.063	0.058	0.068	in		
4	B	Linear Dimension	Caliper	0.625	0.620	0.630	in		
5	B	Linear Dimension	Caliper	0.188	0.183	0.193	in		
6	B	Linear Dimension	Caliper	0.438	0.433	0.493	in		
7	C	Linear Dimension	Caliper	0.063	0.058	0.068	in		
8	C	Linear Dimension	Caliper	0.313	0.308	0.318	in		
9	C	Linear Dimension	Caliper	0.188	0.183	0.193	in		
10	D	Radius	Radius Gauge or Chord Method	5.50	5.48	5.52	in		
11	Notes	Note	Visual	Sharpen Noted Surface	-	-	-		
12	E	Diameter	Caliper	0.064	0.059	0.069	in		
13	E	Depth	Visual	THRU	-	-	-		
14	F	Diameter	Caliper	0.064	0.059	0.069	in		
15	F	Depth	Visual	THRU	-	-	-		
16	Note	Note	Visual	Finish noted surfaces	-	-	-		
17	Note	Note	Visual	Break all edges	-	-	-		
RESULT:								PASS	
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Adam Gulde						☐ Rework	
Group Member 2:		Hayden Patalinghug						☐ Engineering Change	
Group Member 3		Milan Stevanovic		TA Signature					
Group Member 4		Syon Aidt							

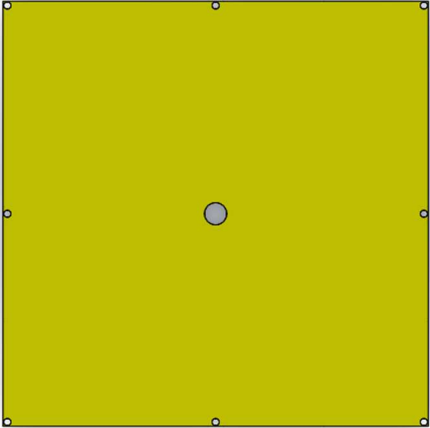
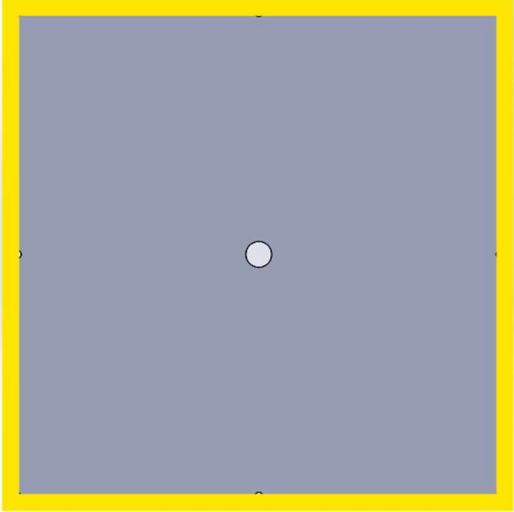
EML 2322L MANUFACTURING INSPECTION RECORD									
Part Number		Revision	Part Name		Team Number		Machinist:		
EML 2322L-P7		A	Blade Holder		7C		Inspector:		
No.	Ref. Location	Feature Type	Measurement Tool/Type	Requirement	Lower Bound	Upper Bound	Units	Results	Comments
1	A	Linear Dimension	Caliper	0.625	0.620	0.630	in		
2	A	Linear Dimension	Caliper	0.313	0.308	0.318	in		
3	B	Linear Dimension	Caliper	0.625	0.620	0.630	in		
4	B	Linear Dimension	Caliper	0.313	0.308	0.318	in		
5	B	Linear Dimension	Caliper	0.188	0.183	0.193	in		
6	B	Linear Dimension	Caliper	0.438	0.433	0.443	in		
7	C	Diameter	Caliper	0.375	0.370	0.380	in		
8	C	Depth	Visual	THRU	-	-	-		
9	D	Linear Dimension	Caliper	0.500	0.495	0.505	in		
10	D	Linear Dimension	Caliper	0.625	0.620	0.630	in		
11	E	Diameter	Caliper	0.064	0.059	0.069	in		
12	E	Depth	Visual	THRU	-	-	-		
13	F	Diameter	Caliper	0.064	0.059	0.059	in		
14	F	Depth	Visual	THRU	-	-	-		
15	Note	Note	Visual	Finish no surfaces	-	-	-		
16	Note	Note	Visual	Break all edges	-	-	-		
RESULT:								PASS	
Inspector Signature:							FAIL	☐ Remanufacture	
Group Member 1:		Adam Gulde						☐ Rework	
Group Member 2:		Hayden Patalinghug						☐ Engineering Change	
Group Member 3		Milan Stevanovic		TA Signature					
Group Member 4		Syon Aidt							

Appendix C: Manufacturing Process Outlines

P1 – Base

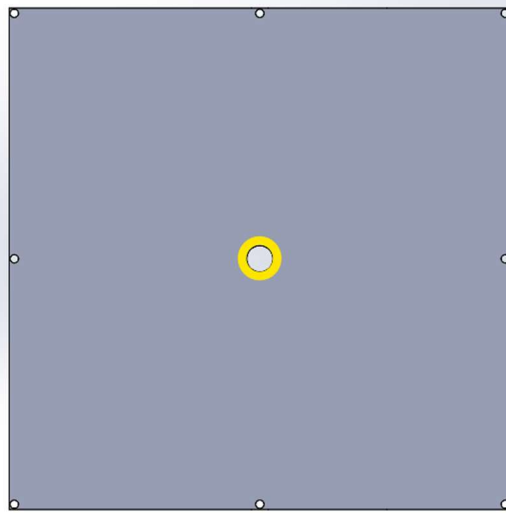
Manufacturing Processes	Process Setup
Setup & Slice 3D Model - Create the part in a CAD software - Export the CAD model as an STL file - Import the STL file into a slicing software to generate the G-code for printing	 A 3D CAD model of a food slicer base. The base is a rectangular box with a central circular hole on the bottom surface. A smaller, separate rectangular component is shown to the left of the main base. The model is rendered in a light gray color.
Setup 3D Printer & Print - Load the G-code to the 3D printer and start printing the part - Remove the part from the build plate once the printing is finished	
Epoxy Resin Application - Apply the epoxy resin to the inner faces of the food slicer area of the base	 A 3D CAD model of the food slicer base, similar to the one in the first row. However, the interior surfaces of the base are highlighted in a bright blue color, indicating the application of epoxy resin. The smaller component to the left is also shown.

P2 – Base Safety Cover

Manufacturing Processes	Process Setup
Material selection & layout - Mark part outline & hole locations on 48" x 48" sheet metal stock	 The diagram shows a large yellow square representing a 48" x 48" sheet metal stock. In the center of the square is a small grey circle representing a hole. At each of the four corners of the square, there is a small grey circle representing a mark for layout.
Sheet metal shearing - Use linear cuts to bring the stock down to the nominal size of 12" x 12"	 The diagram shows a smaller grey square representing a 12" x 12" sheet metal stock. In the center of the square is a small white circle representing a hole. The square is surrounded by a thick yellow border, which represents the linear cuts used to reduce the size of the stock.

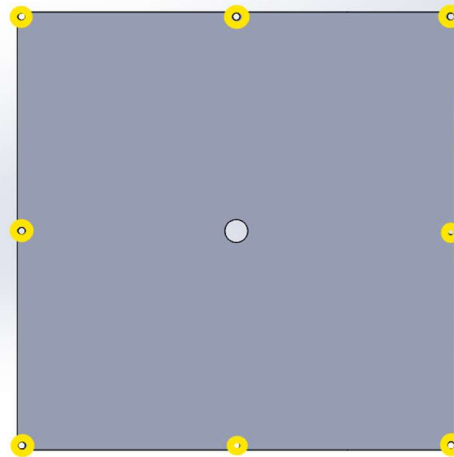
Sheet metal
punching – large
hole

- Use a center
punch and
hammer to prep
the hole
- Use the hole
puncher machine
to punch the
0.625" hole



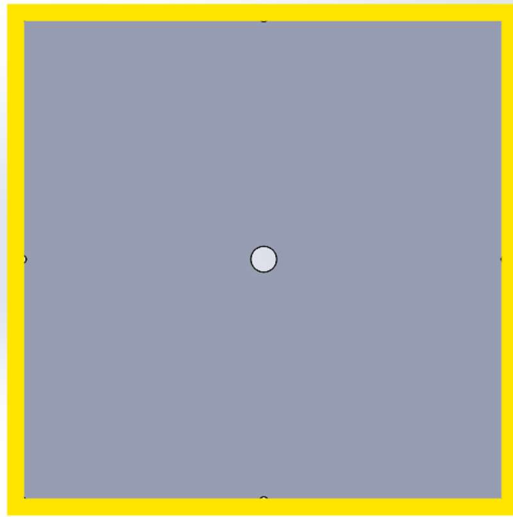
Sheet metal
punching – small
holes

- Use a center
punch and
hammer to prep
the holes
- Use the hole
puncher machine
to punch the
0.196" holes



Deburring /
finishing

- Deburr all
highlighted
edges to remove
sharp edges



P3 – Base Safety Cover

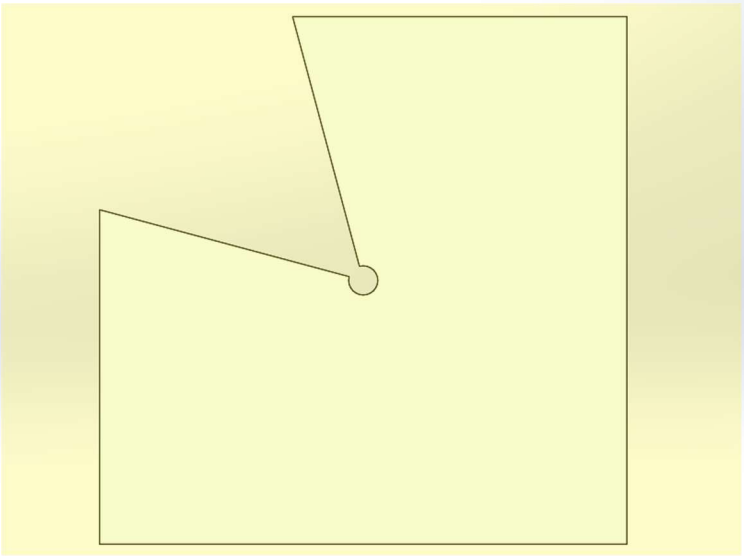
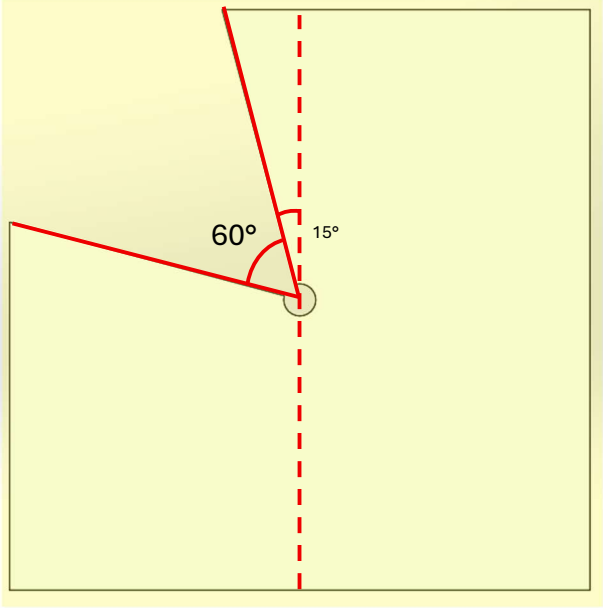
Manufacturing Processes	Process Setup
Material selection & layout - Mark part outline & hole locations on 48" x 48" sheet metal stock	
2. Sheet metal shearing - Use linear cuts to bring the stock down to the nominal size of 6" x 1.25"	
3. Sheet metal punching – holes - Use a center punch and hammer to prep the holes - Use the hole puncher machine to punch the 0.196" holes	

4. Deburring / finishing

- Deburr all highlighted edges to remove sharp edges

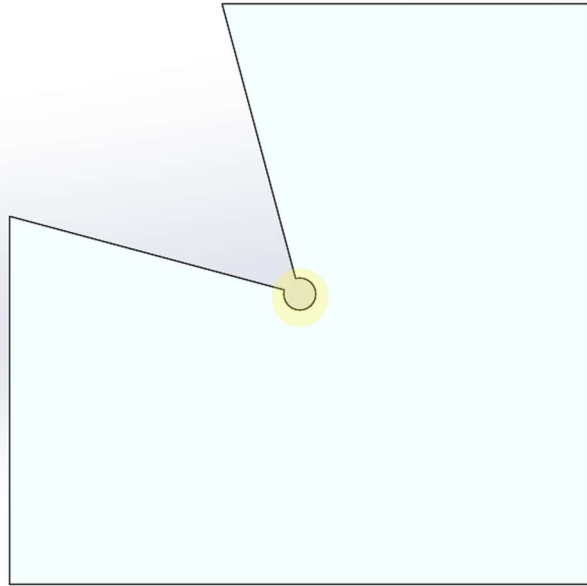


P4 – Plate

Manufacturing Processes	Process Setup
Material Selection and Stock Preparation - Start with a piece of rectangular sheet (thickness no smaller than $0.063 \pm 0.005''$) 6061 aluminum stock	 A diagram of a rectangular plate with a triangular cutout. The cutout is defined by a vertical line on the left, a horizontal line at the top, and a diagonal line connecting the two. A small circle is located at the vertex where the diagonal line meets the horizontal line.
Sheet metal cutting - Using bandsaw, cut stock to square size 11.3"x11.3" - Mark 60° angle onto surface using protractor and straightedge, 15° from a marked vertical centerline. These lines should follow to the center of the plate. - Using bandsaw, cut following the marked lines to create the 60° opening.	 A diagram of a rectangular plate with a 60° cutout. A vertical dashed line represents the centerline. A 60° angle is marked between the centerline and a diagonal line. A 15° angle is marked between the diagonal line and a horizontal line. A small circle is located at the vertex where the diagonal line meets the horizontal line.

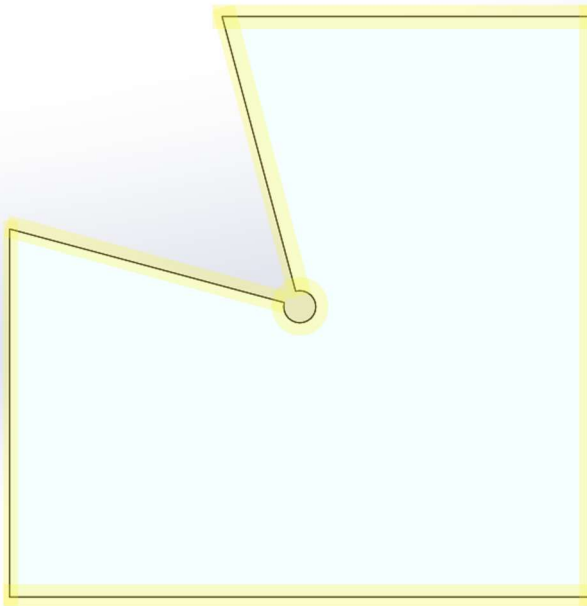
Sheet metal punching

- Using sheet metal hole puncher, punch the 0.625" center hole

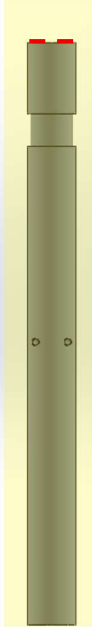


Deburring

- Deburr all highlighted edges to remove sharp edges.



P5 – Shaft

Manufacturing Processes	Process Setup
Material Selection and Stock Preparation - Start with a piece of cylindrical 6061 aluminum stock, diameter 0.375 ± 0.005" and set in a lathe after bandsawing to appropriate length of about 5.5". Set in lathe with 1" from the edge. - Zero on the "top" face and face down to length of 4.50 ± 0.02". (Red dashed line indicates zero face)	
Lathe turning - Zero on the diameter. (red dashed line indicates zero edge) - If stock is not in spec, turn down entire workpiece to larger diameter 0.375". Re-zero. - Turn down smaller diameter region to 0.315". - Include a finishing pass on smaller diameter.	

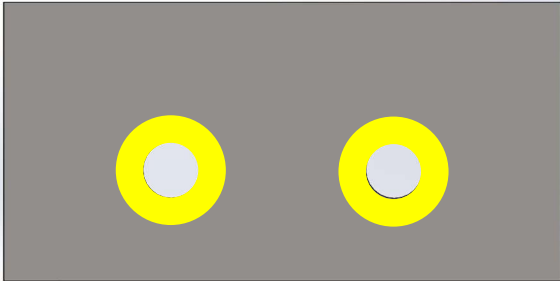
Drilling holes - Use a regular edge finder to zero on the top face for the x-axis. (red dashed line) - Use a cylindrical edge finder to zero on the mill for the y-axis. (yellow dashed lines) - Do a destructive zero on the highest part of the workpiece to set z-axis zero. (anywhere along the blue dashed line) - Use a center drill to peck the hole locations. - Drill with a 0.047" drill to a depth of 0.123".	
Tapping - Using a 0-80 UNC tap in the mill, tap to a depth of 0.063".	
Repeat steps 3-4 for the other side of the workpiece.	

Deburring / finishing

- Move part back to lathe. Rezero using a destructive zero.
- Include a finishing pass on larger diameter region.
- Rezero using a destructive zero on the smaller diameter.
- Include a finishing pass on smaller diameter region.

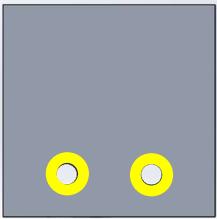
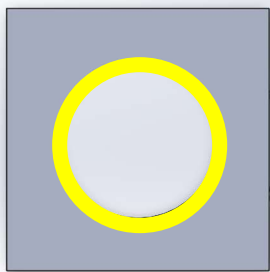


P6 – Blade

Manufacturing Processes	Process Setup
Material Selection and Stock Preparation - Starting with a piece of rectangular 304 stainless steel stock - Using a bandsaw, cut down the stock into a 5" x 0.313" x 1" bar	
Milling off material to create a thin blade - Clamp the stock into the vice with the shown face oriented upwards, use a work stop to avoid having to re-zero, then zero from the right and top faces then use an end mill to remove the material highlighted in yellow, down to 0.625"	
Drilling mounting holes - Rotate the part so that the shown face is oriented upwards, use a center drill at the center of each yellow highlighted hole, then drill out each hole with a 0.064" diameter drill bit.	

Waterjet Cut - Place part into waterjet machine with shown face oriented upwards. Run the waterjet program which will create the curve, as well as sharpening the highlighted portion.	
Deburring - Deburr all edges to remove all sharp edges.	

P7 – Blade Holder

Manufacturing Processes	Process Setup
Material Selection and Stock Preparation - Starting with a piece of rectangular 6061 aluminum stock - Using a bandsaw, mill down the stock to a 0.625" cube	
Drilling first two holes - Clamp the part in the mill with the shown face oriented upwards - Zero from top and left faces, using a workstop to maintain the zero - Use a center drill at the location of each highlighted hole - Drill each to size with a 0.064" diameter drill bit	
Drilling the large top hole - Reorient the part so that the shown face is now upwards - Use a center drill to mark the center of the highlighted hole - Drill with a 0.25" diameter drill bit - Drill with a 0.375" diameter drill bit	
Deburring - Deburr all edges to remove all sharp edges	